Xingyang Li

brucelee_sjtu@sjtu.edu.cn / https://acm.sjtu.edu.cn/~xyli

EDUCATION

Massachusetts Institute of Technology (MIT) — Cambridge, MA
Visiting Student at MIT HAN Lab, EECS Department — 07/2025 – 12/2025
Advisor: Prof. Song Han

Shanghai Jiao Tong University (SJTU) — Shanghai, China
B.Eng. in Computer Science & Technology — 09/2022 – 06/2026 (Expected)
Member of (ACM Honors Class), an elite CS program for top 5% students

- **GPA** (All Core Courses): **4.03**/4.3, Ranking: **3**/29, 19 A+ courses
- Selected core courses: Computer Vision: 100/100, Machine Learning: 97/100, Comprehensive Design for Computer System: 97/100, Information Theory: 98/100, Advanced Compiler Project: 100/100, Linear Algebra: 100/100.

RESEARCH INTERESTS

I am interested in efficient machine learning, with a focus on sparse neural networks and low-bit model quantization. My long-term vision is to help build efficient superintelligence—models that stay compact, accessible, and reliably scalable to serve increasingly broad and demanding real-world needs.

SELECTED PUBLICATIONS

Radial Attention: $\mathcal{O}(n \log n)$ Sparse Attention with Energy Decay for Long Video Generation
Xingyang Li, Muyang Li*, Tianle Cai, Haocheng Xi, Shuo Yang, Yujun Lin, Lvmin Zhang, Songlin Yang, Jinbo Hu, Kelly Peng, Maneesh Agrawala, Ion Stoica, Kurt Keutzer, Song Han*
Annual Conference on Neural Information Processing Systems (NeurIPS), 2025. [arxiv] [code] [website] [blog]

Nebula: Infinite-Scale 3D Gaussian Splatting in VR via Collaborative Rendering and Accelerated Stereo Rasterization
He Zhu, Zheng Liu, Xingyang Li, Anbang Wu, Jieru Zhao, Fangxin Liu, Yiming Gan, Jingwen Leng, Yu Feng
To Appear at International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), 2026; (**Oral Presentation**).

SLTarch: Towards Scalable Point-Based Neural Rendering by Taming Workload Imbalance and Memory Irregularity
Xingyang Li, Jie Jiang*, Yu Feng, Yiming Gan, Jieru Zhao, Zihan Liu, Jingwen Leng, Minyi Guo*
International Conference on Computer-Aided Design (ICCAD), 2025; (**Oral Presentation**). [arxiv]

Harnessing Conventional Video Processing Insights for Emerging 3D Video Generation Models: A Comprehensive Attention-aware Way
Tianlang Zhao, Jun Liu*, Xingyang Li*, Li Ding, Jinhao Li, Shuaiheng Li, Jinbo Hu, Guohao Dai*
Design Automation Conference (DAC), 2025; (**Oral Presentation**). [pdf]

RESEARCH EXPERIENCE

Massachusetts Institute of Technology, MIT HAN Lab — Cambridge, MA
Advisor: Prof. Song Han and Prof. Jun-Yan Zhu

Quantization Algorithm for Efficient Video Diffusion Models — 08/2025 – 11/2025

- Developed **DeltaQuant**, a 4-bit activation–weight quantization method leveraging spatiotemporal similarity for efficient video diffusion deployment. (*first author, submitted to CVPR 2026*)

Efficient and Scalable Video Diffusion Models — 02/2025 – 07/2025

- Proposed **Radial Attention**, a sub-quadratic sparse attention mask enabling longer video generation with substantial training and inference speedups. (*first author, NeurIPS 2025*)

- Open-sourced the system (550+ stars), adopted by state-of-the-art video diffusion inference engines.

Shanghai Jiao Tong University, DAI Lab Shanghai, China
Advisor: Prof. Guohao Dai

Mixed Precision Video Diffusion Transformers with Hardware Accelerator 10/2024 – 12/2024
- Designed block-wise speculative attention and mixed-precision computation for video diffusion acceleration. (*co-first author, DAC 2025*)

Shanghai Jiao Tong University, EPCC Lab Shanghai, China
Advisor: Prof. Yu Feng

Scalable and Low-Latency Mobile Rendering System for 3D Gaussian Splatting 01/2025 – 05/2025
- Developed temporal-aware traversal reuse and kernel prototypes for city-scale mobile 3DGS rendering. (*third author, ASPLOS 2026*)

Algorithm–Hardware Co-design for Scalable 3D Gaussian Splatting 08/2024 – 04/2025
- Proposed a subtree-based traversal algorithm and specialized hardware for scalable 3DGS rendering. (*first author, ICCAD 2025*)

PROJECT EXPERIENCE

Personal Project: Block Sparse Quantized Attention *[repo]* 07/2025 – 08/2025
- Integrated block-sparse APIs into a quantized attention library.
- Pushed the frontier of attention acceleration for video diffusion models.

Course Project: Mx* Compiler with a Just-in-time (JIT) Compiler *[repo]* 07/2023 – 04/2024
- Designed a compiler from C-and-Java-like language to RV32I assembly
- Crafted various LLVM-IR optimizations achieving performance comparable to `-O2`.
- Designed a JIT-Compiler for LLVM.

Course Project: RISC-V CPU implemented in Verilog RTL *[repo]* 10/2023 – 01/2024
- Designed a Tomasulo algorithm-based RISC-V CPU with Write Buffer, ICache, and Branch Prediction.
- Used Vivado to generate bitstream and program the Basys3 FPGA board.

Course Project: Google File System (GFS) *[repo]* 07/2024 – 09/2024
- Implemented the basic functions mentioned in the original GFS Paper in Golang.

TEACHING EXPERIENCE

Machine Learning 02/2025 – 06/2025
Mathematical Logic 02/2024 – 06/2024
Computer Programming 09/2023 – 02/2024

Role as teaching assistant: giving lectures and recitation classes, writing documents and sample solutions, grading homework, creating exam questions, designing ML-related final project lists, etc.

HONORS

2025 Commercial Sponsorship Scholarship (**Top 6** at Zhiyuan College, SJTU)
2025 SenseTime Scholarship (**30 winners nationwide** with aspiration in AI research)
2024 Commercial Sponsorship Scholarship (**14 winners** each year in SJTU)
2023 Longfor Merit Scholarship (**Top 10** at Zhiyuan College, SJTU)

TECHNICAL SKILLS & MISCELLANEOUS

Programming : Python, C++, CUDA, Bash, Git, Java, Go, Verilog
Tools : LaTeX, Markdown, Bash, Git
Language : TOEFL: Total 115 (Reading 30, Listening 30, Speaking 25, Writing 30); CET 6: 665/710
Hobbies : Soccer, Table Tennis, Running, Guitar